

ArogyaRakshak — Literature Bank (Open-Access Sources)

Curated for the Related Work section. All sources below are freely accessible (arXiv, PMC, ACL Anthology, MDPI open-access, or IEEE-listed-but-open). Paywalled sources were deliberately excluded.

1. BillNyay — Overcharge / Fraud Detection

"Fraud detection in healthcare billing and claims" — International Journal of Science and Research Archive, 2024 (open access) https://ijsra.net/sites/default/files/fulltext_pdf/IJSRA-2024-2606.pdf Directly relevant: covers NLP-based detection of discrepancies between narrative descriptions and billed services (upcoding/phantom billing) — the general problem class BillNyay's overcharge detection sits in. Good general framing citation for your Introduction/Related Work, not a specific technique to copy.

"Improving Medicare Fraud Detection Accuracy... Feature Selection and Data Sampling Techniques" — medRxiv, 2026

<https://www.medrxiv.org/content/10.64898/2026.03.18.26348763v1.full> Reports 95.4% accuracy combining feature selection + data sampling for fraud detection; also references a BERT+ICD-code model reaching 87% on suspicious-claim identification. Useful as a benchmark reference point if you build an evaluation section for BillNyay's detection accuracy — gives you comparable numbers from the literature.

"Dialogue to Detection: A Multimodal Hybrid NLP Pipeline for Insurance Fraud

Detection" — arXiv, 2026 <https://arxiv.org/pdf/2606.28002> Combines NER, regex extraction, and LLM-RAG retrieval in a rule-based risk score for insurance fraud — architecturally close to BillNyay's own hybrid rules+RAG+agent design. Good citation for justifying "hybrid rule-based + LLM" as a defensible pattern, not a novelty risk.

Caution across this section: all of these target insurer-side fraud detection (claims fraud), not patient-side overcharge detection (which is what BillNyay actually does). Cite for methodology parallels, not as direct precedent — your problem is the mirror image of theirs (protecting the patient, not the insurer).

2. Kadi — Indic Entity Resolution, Transliteration & NER (directly strengthens your fellowship note)

Naamapadam: A Large-Scale Named Entity Annotated Data for Indic Languages —

arXiv/ACL <https://arxiv.org/pdf/2212.10168> Large-scale NER dataset spanning Indic languages. Directly usable as a starting point or comparison baseline for Kadi's entity extraction, and a strong citation for "Indic NER is an active, resourced research area" in your Related Work.

L3Cube-MahaNER — Litake et al., WILDRE-6 Workshop, LREC 2022 (ACL Anthology)
Cited within the Naamapadam paper above. A Marathi-specific NER dataset + BERT
models from the same L3Cube group whose IndicSBERT model you're already using —
strong continuity citation, same research group, same language.

Aksharantar: Towards building open transliteration tools for the next billion users —
Madhani et al., 2022 Referenced in the Naamapadam bibliography. This is the open
transliteration resource underlying tools in the same family as IndicXlit — good citation for
justifying the transliteration-matching signal in Kadi's entity resolution pipeline.

**"Evaluating Cross-lingual Knowledge Consistency in Code-Mixed vis-a-vis Indian
Languages using IndicKLAR"** — arXiv, 2026 <https://arxiv.org/html/2605.29637> Recent
(2026) work specifically studying failure modes of transliteration-based approaches across
Hindi, Marathi, and other Indic languages — including cases where transliteration
produces phonetically-similar-but-wrong entity names. **This is a genuinely useful citation
for your limitations section** — it gives you literature-backed language for exactly the kind
of transliteration failure mode your entity resolution pipeline needs to handle (medium-
confidence → ask user, rather than auto-merge).

Relevance to your IndiaAI fellowship note: these four together let you write a much
stronger "gap in literature" paragraph — NER and transliteration are separately well-
studied for Indic languages, but a combined confidence-scored entity resolution pipeline
specifically for medical billing documents (your actual research question) isn't covered by
any of them. That's exactly the kind of specific, defensible gap fellowship reviewers want to
see.

3. Hallucination Mitigation / RAG Grounding (justifies BillNyay's QA Judge agent design)

**"Reducing Hallucinations in Medical AI Through Citation Enforced Prompting in RAG
Systems"** — MDPI, 2026 (open access) <https://www.mdpi.com/2076-3417/16/6/3013>
Directly relevant — introduces an "Unsupported Sentence Ratio" metric and citation-
enforced grounding for medical RAG, extremely close to what your QA Judge agent is
already supposed to do (drop unsupported claims). Strong citation to justify that design as
following an established mitigation pattern, not an ad hoc addition. Also flags multilingual
adaptation as future work — worth quoting as evidence that multilingual RAG grounding is
still an open problem (supports your project's novelty).

**"MEGA-RAG: a retrieval-augmented generation framework... for mitigating
hallucinations of LLMs in public health"** — PMC, 2026 (open access)
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12540348/> Public-health-specific RAG
hallucination mitigation with a "Semantic divergence + Evidence coverage" scoring
approach. Good citation for the general pattern of evidence-coverage scoring, which you
could reference when describing the QA Judge's citation-checking logic.

"HalluGraph: Auditable Hallucination Detection for Legal RAG" — arXiv, 2025/2026

<https://arxiv.org/pdf/2512.01659> Legal-domain (not medical) but very relevant to BillNyay's appeal/complaint drafting specifically, since that's a legal-document-generation task.

Reports that even RAG-enhanced systems hallucinate 17–33% of the time on legal content — a genuinely useful, citable number to justify why your mandatory-citation QA Judge step exists, and a realistic baseline to compare your own hallucination rate against if you build an eval for it.

"Mitigating Hallucination in Large Language Models (LLMs): An Application-Oriented Survey on RAG, Reasoning, and Agentic Systems" — arXiv, 2026

<https://arxiv.org/pdf/2510.24476> Broad survey with dedicated sections on both healthcare and legal RAG applications — good single citation to anchor your Related Work's hallucination-mitigation paragraph before drilling into the more specific papers above.

4. ABDM / Digital Health Interoperability (India-specific)

"Awareness of and Challenges in Utilizing the Ayushman Bharat Digital Mission for Healthcare Delivery" — PMC, qualitative study

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11855927/> Real qualitative data on low awareness and data-security concerns as adoption barriers for ABDM — useful for your Introduction/motivation section when framing why a consent-based, user-facing tool (like Kadi's ABDM integration) needs to be simple and trustworthy, not just technically correct.

"A Review on Interoperability Challenges between EMR (Micro) and ABDM (Macro)" — ResearchGate/IRJCS Covers interoperability challenges specifically relevant to why you're scoping ABDM integration to sandbox-only rather than claiming full production interoperability — a citable acknowledgment that this is a recognized, real technical challenge, not just your team's limitation.

Caution: most ABDM literature available openly is qualitative/policy-review rather than technical/benchmarked — expected, since ABDM is infrastructure, not a research artifact. Use these for motivation and context, not for technical methodology citations.

5. OCR — Devanagari / Indic Handwriting (strengthens your OCR validation plan)

Devanagari Handwritten Character Dataset (DHCD) — UCI Machine Learning Repository, open dataset 92,000 grayscale images across 46 classes. Useful as a starting benchmark dataset for validating your OCR pipeline's baseline character-level accuracy on Devanagari before testing on real bill/prescription scans.

IIIT-HW-Dev — "Offline Handwriting Recognition on Devanagari Using a New Benchmark Dataset" — IEEE (openly described on ResearchGate/IEEE abstract), 95,381 word samples from 12 writers <https://ieeexplore.ieee.org/document/8395166/> Word-level

(not just character-level) Devanagari handwriting benchmark — much closer to your real use case (reading full prescriptions/bills) than character-only datasets. Strong citation for your OCR validation methodology.

CHIPS — Corpus of Handwritten Indic Scripts — recent benchmark paper, page-level, 13 Indic languages + English, 15,000 images per language set Page-level (not word/character-level) — the most realistic precedent for validating OCR on full scanned documents like hospital bills. If accessible, this is your strongest citation for justifying a real Devanagari validation test set rather than just claiming "we support Hindi/Marathi OCR."

6. SchemeSetu — RAG for Policy/Legal Eligibility Reasoning

"A Reasoning-Focused Legal Retrieval Benchmark" (Bar Exam QA, Housing Statute QA) — arXiv, 2025 <https://arxiv.org/html/2505.03970v1> Not India-specific, but directly relevant methodologically: benchmarks RAG systems on eligibility-style statute reasoning where query and relevant document have little lexical overlap — exactly SchemeSetu's problem (a user's plain-language situation doesn't lexically match PMJAY/MJPJAY clause wording). Good citation for why SchemeSetu needs semantic retrieval, not keyword matching.

"CBR-RAG: Case-Based Reasoning for Retrieval Augmented Generation in LLMs for Legal Question Answering" — arXiv <https://www.arxiv.org/pdf/2404.04302> Introduces case-based reasoning as a retrieval-structuring method for legal QA. Potentially useful design inspiration for SchemeSetu — instead of pure semantic retrieval, structure retrieval around "similar past eligibility cases," which could improve consistency of scheme-eligibility answers.

"Knowledge Graph Representations for LLM-Based Policy Compliance Reasoning" — arXiv, 2026 <https://arxiv.org/pdf/2604.27713> Reports that knowledge-graph-structured legal information helps LLMs avoid hallucination compared to flat-text RAG. Worth a mention as a possible future-work direction for SchemeSetu (structuring PMJAY/MJPJAY rules as a KG rather than flat retrieved chunks) — you don't need to build this for the FYP, but citing it shows awareness of a stronger approach than what you're shipping, which reads well in a "Future Work" section.

How to use this in your documentation

- Sections 1–3 and 6 are strongest for your **Related Work** chapter — they justify architectural choices (hybrid rules+RAG, citation-enforced generation, semantic retrieval over statutes) with real precedent.
- Section 2 is your **strongest material for the IndiaAI fellowship note** — it sharpens the "genuine gap" argument since NER and transliteration are well-studied separately but not combined for medical entity resolution.

- Section 4 belongs in your **Introduction/motivation**, not Related Work — it's context, not technique.
- Section 5 belongs in your **Evaluation/Methodology** chapter, as the benchmark datasets your OCR validation plan should test against.